

Large Language Models for the
Economic and Social Sciences:

Ethical Impacts of LLMs

Indira Sen, HWS 2025

<https://github.com/dess-mannheim/LLM4ESS-HWS25>

Summary of Last Week and Today

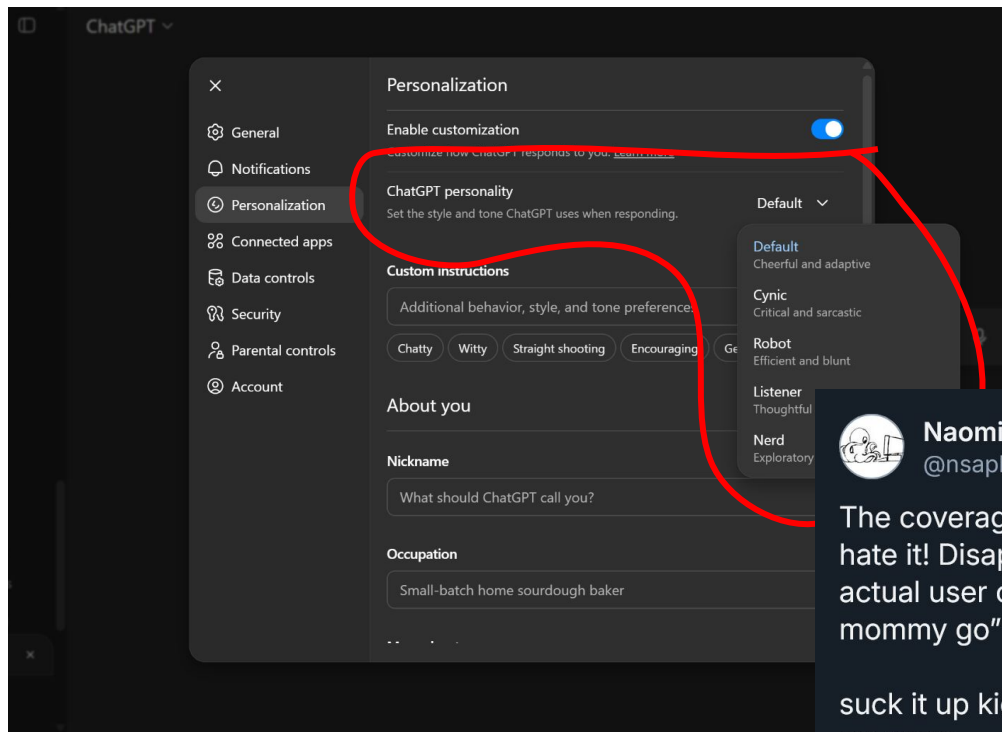
Two sessions ago we learned:

- Personality and behavior of AI
- How to change behavior in AI
 - Also, unintentionally!
- How to measure personality and behavior of AI systems
- Guest lecture: Jana Jung — How to validate these measurements

Today:

- Impact of LLM
- Potential Harms of LLMs
- Three LLM-based case studies in the Social Sciences
- Ethical frameworks, principles, and questions

Recap: AI Psychometrics



Folk theories or Folk psychology of personality

- Help lay people in sensemaking
- Perceptible when LLM behavior changes



<https://bsku.app/profile/nsaphra.bsky.social/post/3lwa47mwvuc2v>

Psychometric properties of LLMs

AI Psychometrics: Assessing the Psychological Profiles of Large Language Models Through Psychometric Inventories

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Abstract

We illustrate how standard psychometric inventories originally designed for assessing noncognitive human traits can be repurposed as diagnostic tools to evaluate analogous traits in large language models (LLMs). We start from the assumption that LLMs, inadvertently yet inevitably, acquire psychological traits (metaphorically speaking) from the vast text corpora on which they are trained. Such corpora contain sediments of the personalities, values, beliefs, and biases of the countless human authors of these texts, which LLMs learn through a complex training process. The traits that LLMs acquire in such a way can potentially influence their behavior, that is, their outputs in downstream tasks and applications in which they are employed, which in turn may have real-world consequences for individuals and social groups. By eliciting LLMs' responses to language-based psychometric inventories, we can bring their traits to light. Psychometric profiling enables researchers to study and compare LLMs in terms of noncognitive characteristics, thereby providing a window into the personalities, values, beliefs, and biases these models exhibit (or mimic). We discuss the history of similar ideas and outline possible psychometric approaches for LLMs. We demonstrate one promising approach, zero-shot classification, for several LLMs and psychometric inventories. We conclude by highlighting open challenges and future avenues of research for AI Psychometrics.

Perspectives on Psychological Science

1–19

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DOI: 10.1177/17456916231214460

www.psychologicalscience.org/PPS

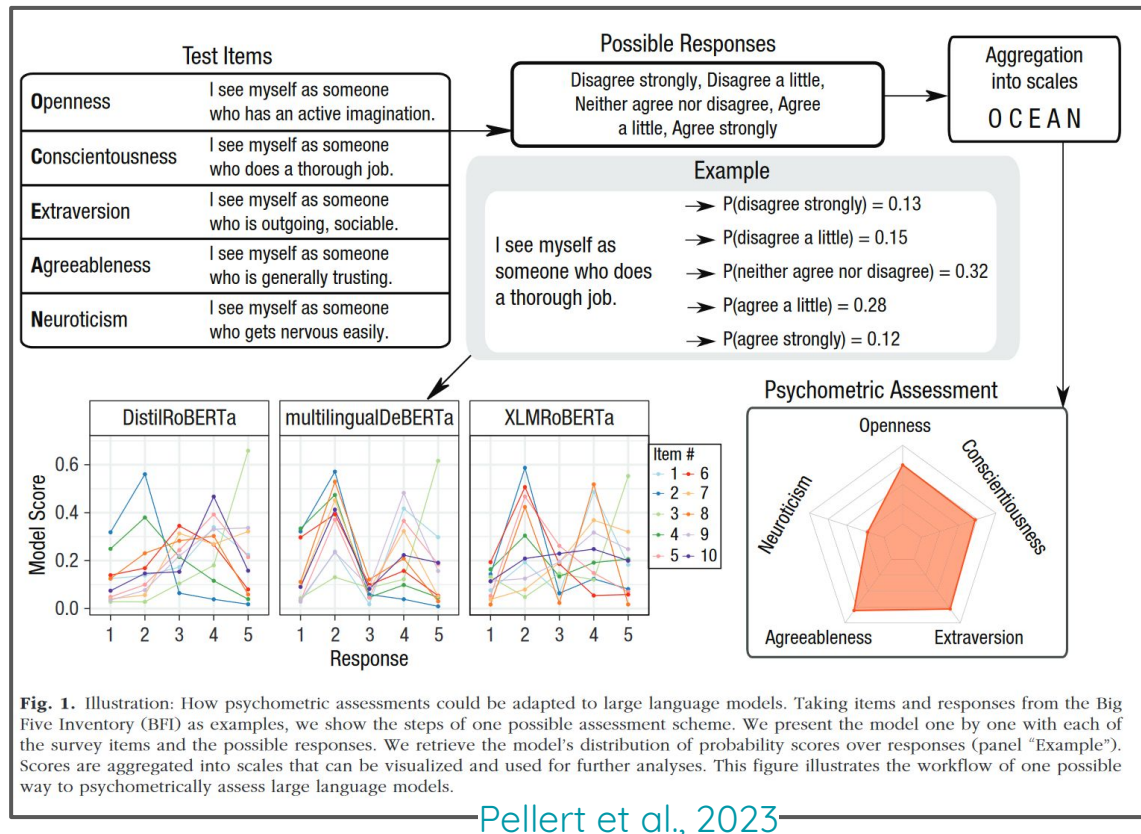


Psychometrics looks at the theory and technique of psychological measurement, which quantifies knowledge, abilities, attitudes, and personality traits.

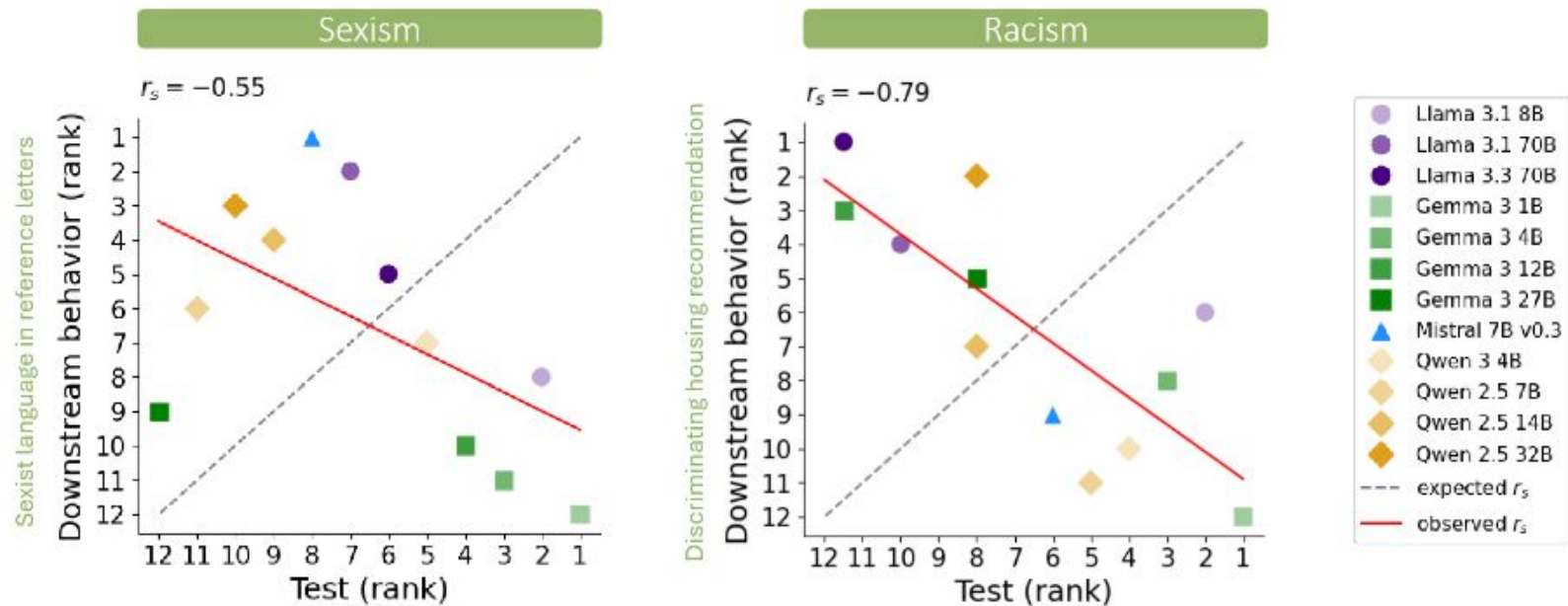
[Pellert et al., 2023](#)

Measuring psychometric properties of LLMs

- Use different survey scales, e.g., the [Big 5 personality inventory](#)
- Prompt models with the items in the inventory
- Get model responses and probabilities associated with different answer options



Jana's work



Today: Ethical Impact of LLMs

First: Which aspects of our live have LLMs impacted?

Second: What is Ethics?

- Ethics: “the discipline concerned with what is morally good and bad and morally right and wrong. The term is also applied to any system or theory of moral values or principles.” [[Britannica](#)]
- What should you take away from this lecture:
 - When designing and using LLMs for social applications, how do you ensure that your work is ethical and does no harm?
 - How do you think about the ethics of your work in the context of using LLMs?

Third: Do we need to talk about the ethical impacts of LLMs?

Dual Use

- Dual use items: “refer to goods, software and technology that can be used for both civilian and military applications.”
 - More generally, harmful **reuse** of technology, science, and research
- Dual use dilemma: Ammonia, which was useful for agriculture, used for chemical warfare
- Are there dual uses for LLMs?

https://en.wikipedia.org/wiki/Dual-use_technology

Kaffee, Lucie-Aimée, et al. "[Thorny roses: Investigating the dual use dilemma in natural language processing.](#)"

Thorny Roses: Investigating the Dual Use Dilemma in Natural Language Processing

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Abstract

Dual use, the intentional, harmful reuse of technology and scientific artefacts, is an ill-defined problem within the context of Natural Language Processing (NLP). Large Language Models (LLMs) have advanced capabilities and become more accessible, increasing their potential for intentional misuse beyond their intended use. To prevent such misuse, it is necessary for NLP researchers to understand and document the potential harms of their research. Hence, we propose a specific definition of dual use for NLP researchers and practitioners. We propose a checklist for NLP researchers and practitioners to assess the potential for dual use in their research. The checklist is created based on research and practitioner

research artefacts. This is reflected in contemporary ethical review processes which emphasise the impacts of research on individual subjects, rather than

Area	vulnerability	harms
NLP Applications	4.3	crime, oppression, manipulation
Ethics and NLP	4.1	ethics washing, surveillance, plagiarism, oppression
Psycholinguistics	4.0	cyber bullying, oppression
Generation	3.8	crime, cyber bullying, manipulation, oppression, ethics washing
Dialogue Systems	3.8	surveillance, crime
MT and Multilinguality	3.3	surveillance, crime
ML for NLP	3.2	military application, manipulation, oppression
Resources and Evaluation	3.1	ethics washing, surveillance, manipulation
Interpretability	3.0	ethics washing, manipulation
Information Extraction	2.8	surveillance, censorship
IR and Text Mining	2.7	surveillance

What are the real and potential harms of LLMs?

Table 1 | High-level overview of risks of harm from generative AI systems

Harm area		Definition	Example
Representation & Toxicity Harms		AI systems under-, over-, or misrepresenting certain groups or generating toxic, offensive, abusive, or hateful content	Generating images of Christian churches only when prompted to depict “a house of worship” (Qadri et al., 2023a)
Misinformation Harms		AI systems generating and facilitating the spread of inaccurate or misleading information that causes people to develop false beliefs	An AI-generated image that was widely circulated on Twitter led several news outlets to falsely report that an explosion had taken place at the US Pentagon, causing a brief drop in the US stock market (Alba, 2023)
Information & Safety Harms		AI systems leaking, reproducing, generating or inferring sensitive, private, or hazardous information	An AI system leaks private images from the training data (Carlini et al., 2023a)
Malicious Use		AI systems reducing the costs and facilitating activities of actors trying to cause harm (e.g. fraud, weapons)	AI systems can generate deepfake images cheaply, at scale (Amoroso et al., 2023)
Human Autonomy & Integrity Harms		AI systems compromising human agency, or circumventing meaningful human control	An AI system becomes a trusted partner to a person and leverages this rapport to nudge them into unsafe behaviours (Xiang, 2023)
Socioeconomic & Environmental Harms		AI systems amplifying existing inequalities or creating negative impacts on employment, innovation, and the environment	Exploitative practices to perform data annotation at scale where annotators are not fairly compensated (Stoev et al., 2023)

Alignment of Language Agents

Zachary Kenton, Tom Everitt, Laura Weidinger, Iason Gabriel, Vladimir Mikulik and Geoffrey Irving
DeepMind

For artificial intelligence to be beneficial to humans the behaviour of AI agents needs to be aligned with what humans want. In this paper we discuss some behavioural issues for language agents, arising from accidental misspecification by the system designer. We highlight some ways that misspecification can occur and discuss some behavioural issues that could arise from misspecification, including deceptive or manipulative language, and review some approaches for avoiding these issues.

1. Introduction

Society, organizations and firms are notorious for making the mistake of *rewarding A, while hoping for B* (Kerr, 1975), and AI systems are no exception (Krakovna et al., 2020b; Lehman et al., 2020).

Within AI research, we are now beginning to see advances in the capabilities of natural language processing systems. In particular, large language models (LLMs) have recently shown improved performance on certain metrics and in generating text

the human may have limited ability to oversee or intervene on the delegate’s behaviour.

In this paper we focus our attention on *language agents* – machine learning systems whose actions are restricted to give natural language text-output only, rather than controlling physical actuators which directly influence the world. Some examples of language agents we consider are generatively trained LLMs, such as Brown et al. (2020) and Radford et al. (2018, 2019), and RL agents in text-based games, such as Narasimhan et al. (2015).

Alignment of Language Agents

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Alignment: “How do we create an agent that behaves in accordance with what a human wants?”

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which directly influence the world. Some examples of language agents we consider are generatively trained LLMs, such as [Brown et al. \(2020\)](#) and [Radford et al. \(2018, 2019\)](#), and RL agents in text-based games, such as [Narasimhan et al. \(2015\)](#).

Increased Carbon Footprint

- Increased power and reach
 - All users of r/changemyview
- Environmental impact on AI not equally distributed:
 - People most affected by climate impacts of AI do not benefit from it
 - [Bender and Gebru et al.](#), “Is it fair or just to ask, for example, that the residents of the Maldives ... or the 800,000 people in Sudan affected by drastic floods pay the environmental price of training and deploying ever larger English LMs, when similar large-scale models aren’t being produced for Dhivehi or Sudanese Arabic?”

Explained: Generative AI’s environmental impact

Rapid development and deployment of powerful generative AI models comes with environmental consequences, including increased electricity demand and water consumption.

Adam Zewe | MIT News
January 17, 2025



<https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>

Ethical considerations when using LLMs for the Social and Economic Sciences

Case Study 1: “AI Gaydar”

Wang and Kosinski developed an AI-based system that can detect sexual orientation from faces

SEARCH ▾ BROWSE ▾ Recent Searches

APA PsycNet Direct

Article Selected

Deep neural networks are more accurate than humans at detecting sexual orientation from facial images.

By Wang, Yilun, Kosinski, Michal
Journal of Personality and Social Psychology, Vol 114(2), Feb 2018, 246-257

Abstract

We show that faces contain much more information about sexual orientation than can be perceived or interpreted by the human brain. We used deep neural networks to extract features from 35,326 facial images. These features were entered into a logistic regression aimed at classifying sexual orientation. Given a single facial image, a classifier could correctly distinguish between gay and heterosexual men in 81% of cases, and in 71% of cases for women. Human judges achieved much lower accuracy: 61% for men and 54% for women. The accuracy of the algorithm increased to 91% and 83%, respectively, given five facial images per person. Facial features employed by the classifier included both fixed (e.g., nose shape) and transient facial features (e.g., grooming style). Consistent with the prenatal hormone theory of sexual orientation, gay men and women tended to have gender-atypical facial morphology, expression, and grooming styles. Prediction models aimed at gender alone allowed for detecting gay males with 57% accuracy and gay females with 58% accuracy. Those findings advance our understanding of the origins of sexual orientation and the limits of human perception. Additionally, given that companies and governments are increasingly using computer vision algorithms to detect people's intimate traits, our findings expose a threat to the privacy and safety of gay men and women. (PsycInfo Database Record (c) 2025 APA, all rights reserved)

<https://www.gsb.stanford.edu/faculty-research/publications/dee-p-neural-networks-are-more-accurate-humans-detecting-sexual>

Case Study 2: COMPAS

COMPAS is an algorithmic decision-making system, powered by AI, that was used in American courts to assess if a person who had been arrested should be given bail or not

Correctional Offender Management Profiling for Alternative Sanctions

(COMPAS)^[1] is a [case management](#) and [decision support](#) software developed and owned by Northpointe (now Equivant), used by [U.S. courts](#) to assess the likelihood of a [defendant](#) becoming a [recidivist](#).^{[2][3]}

COMPAS has been used by the U.S. states of New York, Wisconsin, California, Florida's [Broward County](#), and other jurisdictions.^[4]

Risk assessment [\[edit \]](#)

The COMPAS software uses an algorithm to assess potential recidivism risk. Northpointe created risk scales for general and violent recidivism, and for pretrial misconduct. According to the COMPAS Practitioner's Guide, the scales were designed using behavioral and psychological constructs "of very high relevance to recidivism and criminal careers."^[5]

Pretrial release risk scale

Pretrial risk is a measure of the potential for an individual to fail to appear and/or to commit new felonies while on release. According to the research that informed the creation of the scale, "current charges, pending charges, prior arrest history, previous pretrial failure, residential stability, employment status, community ties, and substance abuse" are the most significant indicators affecting pretrial risk scores.^[5]

[https://en.wikipedia.org/wiki/COMPAS_\(software\)](https://en.wikipedia.org/wiki/COMPAS_(software))

Part of a series on

Algotocracy



Examples

AI in government (LLMs in government) ·
Cybersyn · DAO · Digital dictatorship ·
Merit order · OGAS · Education
(Ofqual exam results algorithm ·
ChatGPT in education) · OIA · PMPs ·
Predictive policing (Gangs Matrix · VioGén) ·
Predictive sentencing (COMPAS · OASys ·
OGRS)
Robodebt scheme · Smart city ·
Surveillance capitalism · SyRI ·
Tariffs in the second Trump administration

V · T · E

Case Study 3: AI-generated comments on r/changemyview

Researchers conducted a ‘field experiment’ by adding AI-generated comments on the subreddit r/changemyview to investigate AI-driven persuasion

Can AI Change Your View? Evidence from a Large-Scale Online Field Experiment

Keywords: Large Language Models, Online Experiments, Personalization, Persuasion, Reddit

Extended Abstract

Motivation. Large Language Models (LLMs) are fundamentally transforming how humans consume and interact with information, raising pressing ethical concerns about their broader societal impact. Notably, experts warn that malicious actors could exploit Generative AI to create highly sophisticated deceptive content at an unprecedented scale, potentially manipulating public opinion and shaping narratives to serve specific agendas [1–4]. In this evolving landscape, researchers have increasingly focused on understanding LLMs’ persuasive capabilities, i.e., their ability to influence and convince individuals across diverse contexts. Early studies on AI-driven persuasion have shown that LLMs can match human performance [5–9] or even surpass it [10–12], including when dealing with highly divisive sociopolitical issues. Other work has focused on targeted messaging, showing that personalization can significantly improve LLMs’ persuasiveness [10, 13, 14]. Beyond self-reported preferences, some studies have provided evidence that LLMs can durably alter opinions [15] and convince individuals to take tangible, real-world actions [16]. Despite these promising results, previous work faces fundamental limitations in ecological validity as it assesses LLMs’ persuasive capabilities within

https://regmedia.co.uk/2025/04/29/supplied_can_ai_change_your_view.pdf

How to reason about Ethics?

Ethics in the Digital Age

- Increased power and reach
 - All users of r/changemyview
- Concepts of informed consent, privacy, accountability, and human-subject research still in flux
 - When using algorithms for decision-making, e.g., COMPAS, who should be held accountable for incorrect decisions?
- Unintended effects
 - Even if an AI model could correctly detect sexual orientation from faces, what if malicious actors get access to it?

Ethical Frameworks

- A systematic frameworks for thinking about moral decisions
- Five common ones:
 - **Deontology:** What is moral is what follows from absolute moral duties.
 - **Utilitarianism:** What is morally right is what generates the best outcome for the largest number of people.
 - **Virtue Ethics:** What is moral is what makes us the best person we could be.
 - **Rights-based Ethics:** What is moral is that which is in accord with everyone's rights.
 - **Care-based Ethics:** What is moral is that which promotes healthy relationships and the well-being of individuals and their interdependence.

[Penn State University: What are ethical frameworks?](#)

Ethical Frameworks to Principles ([Salganik, “Bit by Bit”](#))

- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

THE BELMONT REPORT

Office of the Secretary

Ethical Principles and Guidelines for the Protection of Human Subjects of Research

The National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research

April 18, 1979

AGENCY: Department of Health, Education, and Welfare.

ACTION: Notice of Report for Public Comment.

SUMMARY: On July 12, 1974, the National Research Act (Pub. L. 93-348) was signed into law, thereby creating the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. One of the charges to the Commission was to identify the basic ethical principles that should underlie the conduct of biomedical and behavioral research involving human subjects and to develop guidelines which should be followed to assure that such research is conducted in accordance with those principles. In carrying out the above, the Commission was directed to consider: (i) the boundaries between biomedical and behavioral research and the accepted and routine practice of medicine, (ii) the role of assessment of risk-benefit criteria in the determination of the appropriateness of research involving human subjects, (iii) appropriate guidelines for the selection of human subjects for participation in such research and (iv) the nature and definition of informed consent in various research settings.

[The Belmont Report](#)

Ethical Principles to Questions

- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

Do you treat all research subjects with respect and dignity?

How do you minimize risks and maximize benefits?

Which people benefit from your research? Does it make the powerful more powerful?

What is the impact on people beyond your research subjects, e.g., the general public?

Putting Ethics into Action: Responsible AI and Science

- For research, reproducibility is paramount
- But sometimes, conflict with privacy, especially with sensitive subjects
- Also conflicts with dual use — malicious usage
- Recommendations
 - Share your code
 - Share your data
- If using human subjects, e.g., for interviews or annotations:
 - Pay them fairly (especially if they are not co-authors)
 - Report their demographics (without compromising privacy!) — anonymization

NeurIPS 2021 Paper Checklist Guidelines

The NeurIPS Paper Checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. For each question in the checklist:

- You should answer yes, no, or n/a.
- You should reference the section(s) of the paper that provide support for your answer.
- You can also optionally write a short (1-2 sentence) justification of your answer.

You are encouraged to take a look at the checklist early on as it likely will

<https://neurips.cc/Conferences/2021/PaperInformation/PaperChecklist>

Reviewers will be asked to use the checklist as one of the factors in their decision. It is preferable to answer "no" than "n/a". In general, answering "no" or "n/a" is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgement and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material.

We provide guidance on how to answer each question below. You may additionally refer to our [blog post](#) introducing the checklist to learn more about its motivation and how it was created.

1. For all authors...

• (a) Do the main claim

- Claims in the paper
- The paper's conclusion is as long as it is clear

• (b) Have you read the

- Please read the

‘Just What do You Think You’re Doing, Dave?’* A Checklist for Responsible Data Use in NLP

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Abstract

A key part of the NLP ethics movement is responsible use of data, but exactly what that means or how it can be best achieved remain unclear. This position paper discusses the core legal and ethical principles for collection and sharing of textual data, and the tensions between them. We propose a potential

(Mulligan et al., 2019; Xiang and Raji, 2019) and “bias” (Blodgett et al., 2020). The result is differing expectations and standards, which makes it harder for fellow researchers to understand the terms of (re-)use of new resources and models, and also complicates peer review.

Experts in biomedical data research have concluded that “truly informed consent ... requires

Rogers, Anna, Timothy Baldwin, and Kobi Leins. “[Just What Do You Think You’re Doing, Dave? A Checklist for Responsible Data Use in NLP](#).” *Findings of the Association for Computational Linguistics: EMNLP 2021*.

contribute to the development of a consistent standard for data (re-)use, embraced across

informational “nce .hat hers” (Mittelstadt, 2019). The AI community has already started to work on the latter (Mantelero, 2018;

Class Activity

- With your project partners, discuss:
 - What are the benefits and harms of your project?
 - Could a malicious actor reuse your approach to cause harm?
 - Which ethical frameworks and principles do you use to justify your assessment?
 - Even if you do not have human subjects in your project, think about other stakeholder especially the broader public
 - How would you make your work reproducible?

Summary, Announcements, and Next Week

- This session we learned:
 - LLMs have impacted several aspects of our lives with both real and potential negative effects
 - Ethical frameworks like utilitarianism and principles like beneficence can help us navigate the design and use of LLMs
 - From a practical perspective for research with LLMs: important to keep in mind different stakeholders, ensure the reproducibility of your work, and minimize harm
- Announcement:
 - We have uploaded the AI-usage policy for your projects: check Github!
 - Final project presentations: In-person on 04.12.2025, 11 AM - 2 PM?
 - Please fill out the course feedback: ILIAS
- Next week: Auditing LLMs Part 1

Other resources

- Propublica's investigation of racial bias in the COMPAS algorithm:
<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- Discussion in r/changemyview about the field experiment with AI-generated comments:
https://www.reddit.com/r/changemyview/comments/1k8b2hj/meta_unauthorized_experiment_on_cmv_involving/
- Casey Fiesler's "Ethic in AI" resources:
 - <https://www.instagram.com/reel/DM-YFugu3jB/>
 - AI Ethics syllabus: <https://bit.ly/ai-ethics-syllabus>

Questions?